

Does OPS Predict Wins?

A statistical investigation into offense, run prevention, and team success in the 2026 MLB season

Prepared by Akshaya Mamidipalli | Python · pandas · matplotlib · scikit-learn

Note on data coverage: figures reflect data collected roughly midway through the 2026 season (approximately 104–106 games per team) and should be read as an in-progress indicator rather than a final, end-of-season result.

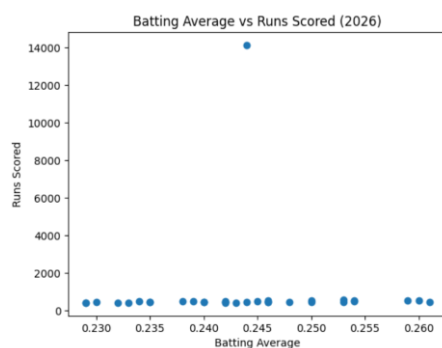
Executive Summary

On-base Plus Slugging (OPS) is the strongest single predictor of team runs scored among standard batting statistics ($r = 0.88$). However, its predictive power drops sharply when applied to team wins ($R^2 = 0.169$), since wins depend on both scoring and preventing runs. This report documents the full analytical process — data collection, cleaning, single- and multi-variable modeling — culminating in a combined model that explains 76% of the variation in team wins.

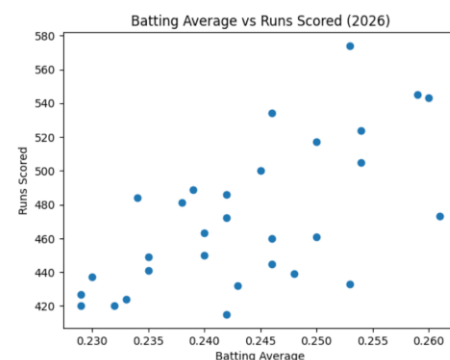
Data Sources & Methodology

Two 2026-season datasets were retrieved from Baseball-Reference.com: Team Standard Batting (AVG, OBP, SLG, OPS, Runs Scored) and Expanded Standings (Wins, Runs Allowed), each covering all 30 teams. Both were exported as CSV and processed in Python using pandas, matplotlib, and scikit-learn, then merged on team name for the wins-based analysis.

Raw exports include a “League Average” row and a combined totals row, neither representing an actual team. Left uncleaned, the totals row alone (summing all 30 teams’ runs) produced a false outlier exceeding 14,000 runs, compressing every real team into a flat line. Both rows were identified and removed prior to analysis:

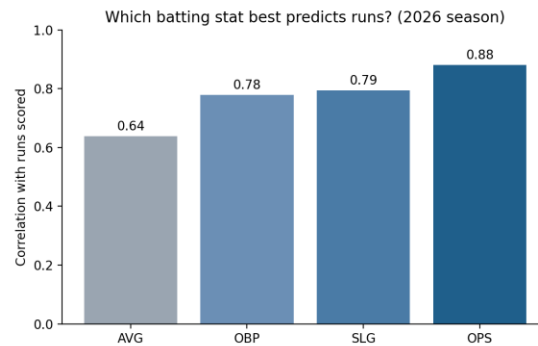


Before: uncleaned totals row



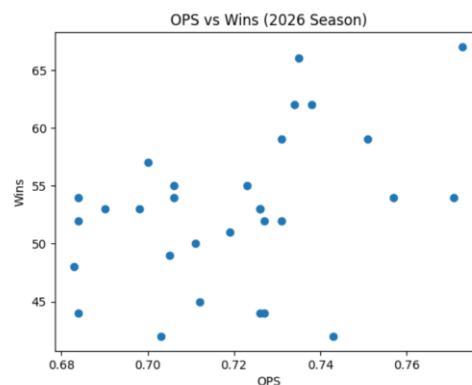
After: League Average and totals rows removed

Finding 1 — OPS Best Predicts Runs Scored



Across four standard batting statistics, OPS produced the strongest correlation with Runs Scored ($r = 0.88$), ahead of SLG (0.79), OBP (0.78), and AVG (0.64) — consistent with the sabermetric principle that reaching base and hitting for power together explain scoring better than batting average alone.

Finding 2 — OPS Alone Weakly Predicts Wins



Following the batting–standings merge, OPS vs. Wins showed a markedly noisier relationship: $r = 0.41$, $R^2 = 0.169$. As a purely offensive measure, OPS does not account for pitching or run prevention, both of which directly influence win totals.

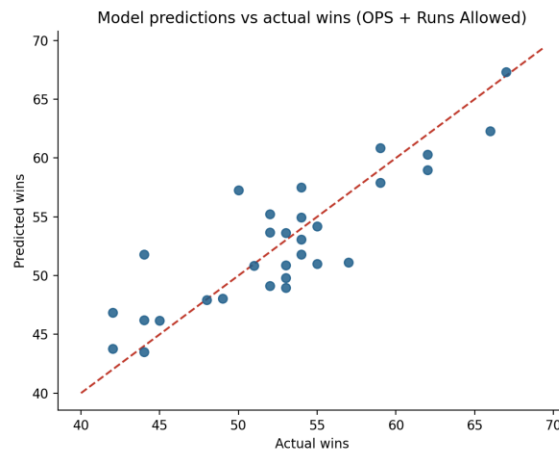
Finding 3 — Selecting a Second Predictor

Two candidates were evaluated as a second predictor: Run Differential ($r = 0.91$ with Wins) and Runs Allowed ($r = -0.71$). Run Differential was excluded despite its higher correlation, as it is derived directly from Runs Scored minus Runs Allowed and is therefore close to a restatement of winning itself. Runs Allowed was selected, as it independently captures pitching and defensive performance without overlapping conceptually with OPS.

Finding 4 — Combined Model

$$\text{Wins} = 132.57 \times \text{OPS} - 9.33 \times \text{RA} - 0.63$$

Combining OPS and Runs Allowed raised R^2 to 0.756 — explaining 76% of win variation, versus 17% for OPS alone. Predicted and actual wins align closely for most teams; the largest deviations (San Francisco, Detroit below prediction; Philadelphia, Minnesota, Arizona above) likely reflect factors outside the model, such as bullpen performance or close-game results.



Conclusion

OPS is a strong indicator of offensive output but an incomplete predictor of team success in isolation. Incorporating run prevention alongside offense substantially improves predictive accuracy, reinforcing that winning in baseball reflects performance on both sides of the ball. As noted above, these results reflect a partial 2026 season and should be interpreted accordingly.

Tools & Data Sources

- Python, pandas, matplotlib, scikit-learn
- Data: Baseball-Reference.com, 2026 MLB season (collected mid-season)
- Notebook and data: github.com/AkshayaMamidipalli/Baseball-Analytics